

Quantum compilation and resource estimation

(II) FTQC resource estimation

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AlgoLab Summer School – Monday 15 June 2026



THE UNIVERSITY OF BRITISH COLUMBIA
Electrical and Computer Engineering
Faculty of Applied Science



Stewart Blusson
Quantum Matter Institute
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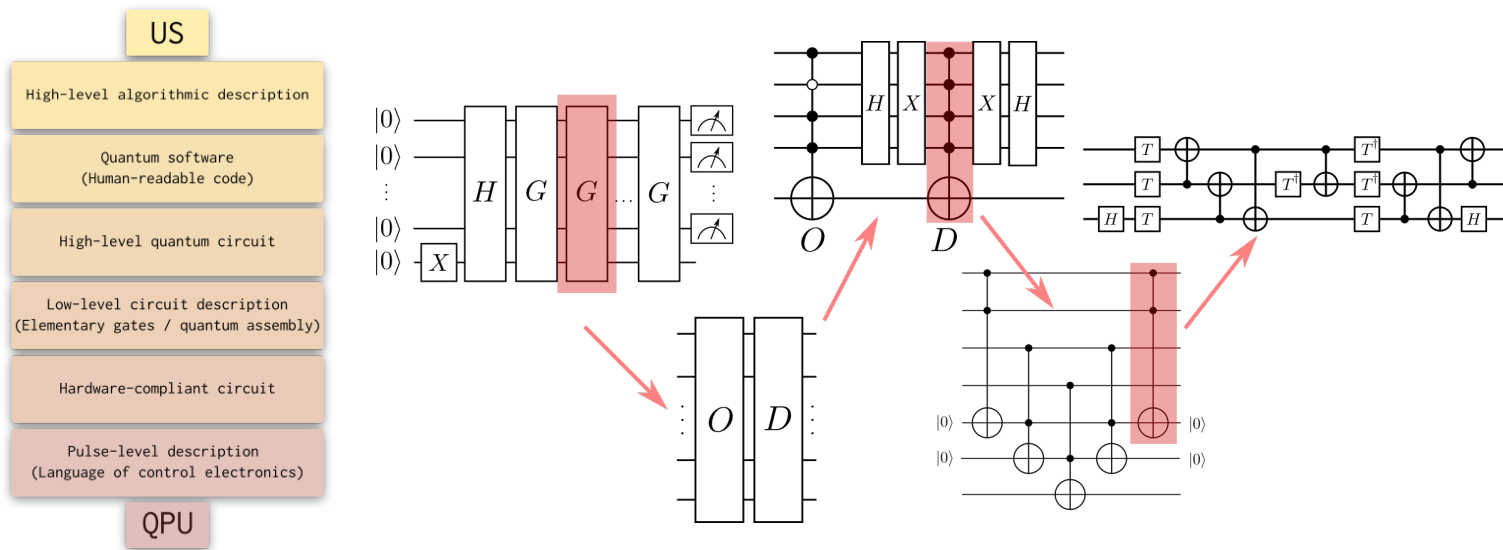


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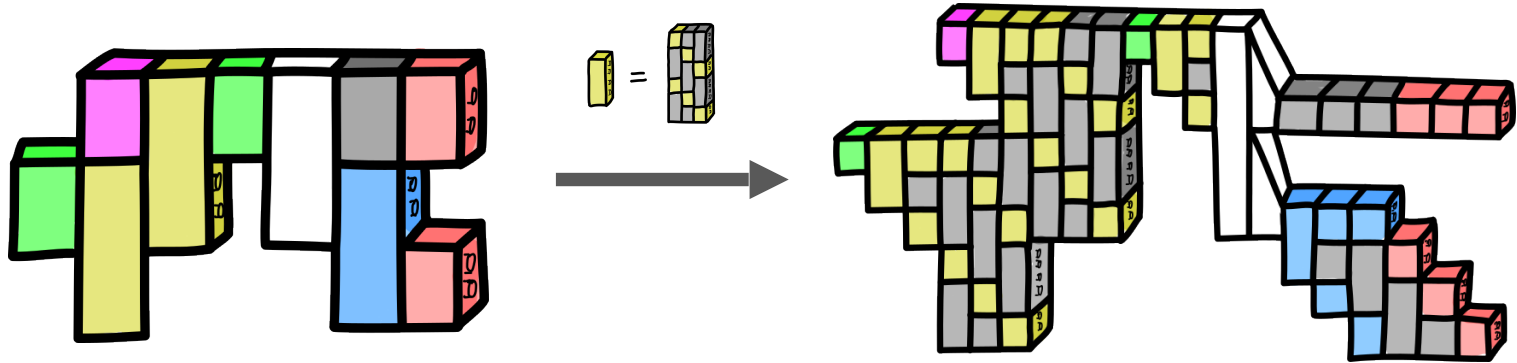
Recap



What are the relevant costs when we add error correction?

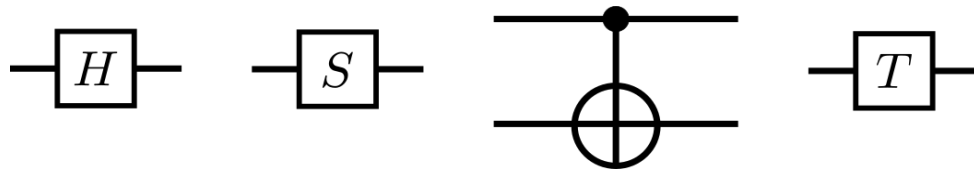
Lecture 2 in one slide

Carry out a “back-of-the-envelope” resource estimate, because things get messy.



What about error correction?

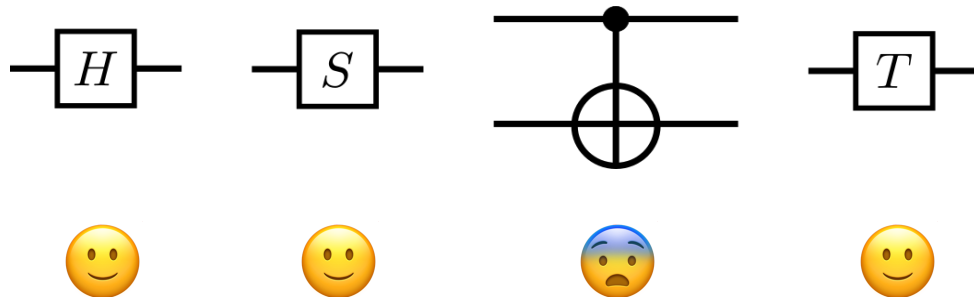
Compilation and optimization is with respect to different metrics and constraints.



What about error correction?

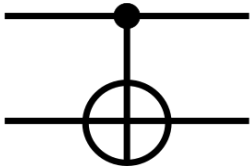
Compilation and optimization is with respect to different metrics and constraints.

NISQ



What about error correction?

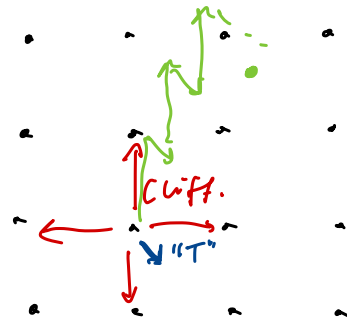
Compilation and optimization is with respect to different metrics and constraints.

				
NISQ				
FTQC				

What about error correction?

Gate count for $RZ(\pi/3)$

- NISQ: RZ (1)
- FTQC: H (49), S (27), T (48), Z (1) (*encoded in a QECC*)

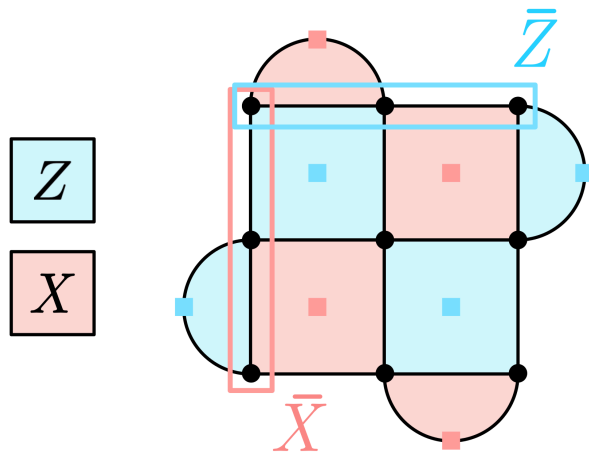


```
—S—H—T—S—H—T—S—H—T—S—H—T—S—H—T—H—T—H—T—S—H—T—S—H—T—H—T—S—H—T—S ...  
... —H—T—H—T—H—T—S—H—T—H—T—S—H—T—H—T—S—H—T—S—H—T—S—H—T—H—T—H—T ...  
... —H—T—S—H—T—H—T—H—T—S—H—T—H—T—S—H—T—H—T—H—T—H—T—H—T—H—T—S—H ...  
... —T—H—T—H—T—S—H—T—H—T—S—H—T—H—T—S—H—T—S—H—T—S—H—T—S—H—T—S—H ...  
... —T—S—H—T—S—H—Z—GlobalPhase(2.36)— Probs
```

What about error correction?

Maintaining fault-tolerance and code properties imposes additional constraints on circuit design and scheduling.

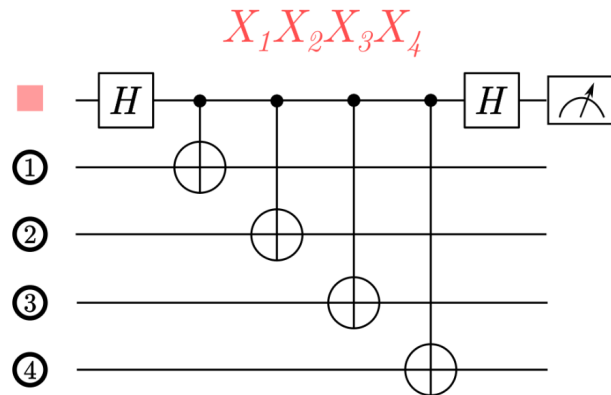
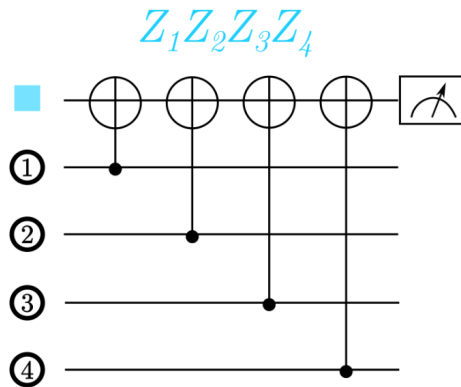
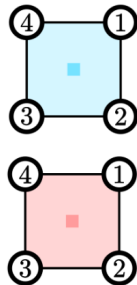
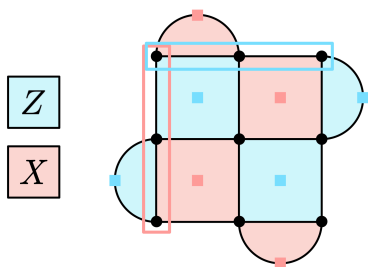
Example: surface code syndrome measurement



What about error correction?

Maintaining fault-tolerance and code properties imposes additional constraints on circuit design and scheduling.

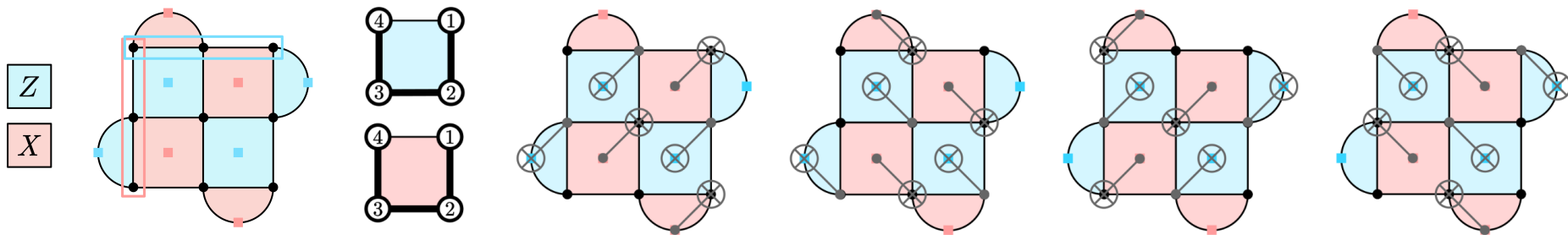
Example: surface code syndrome measurement



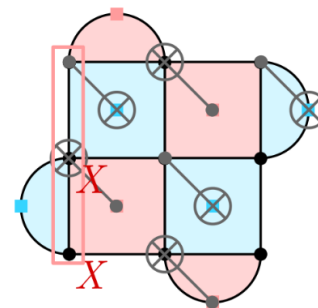
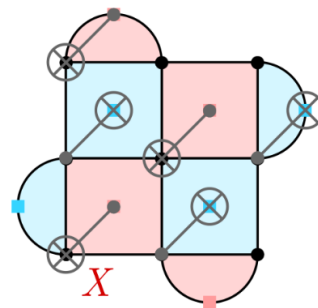
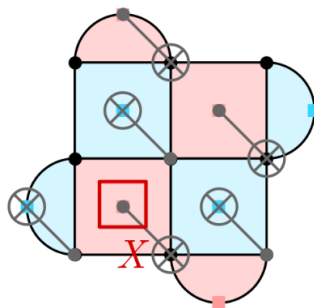
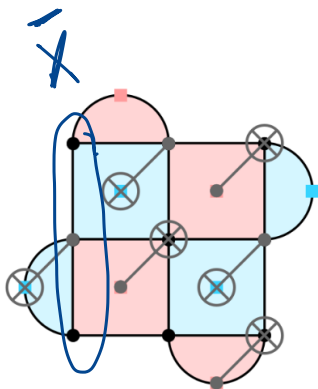
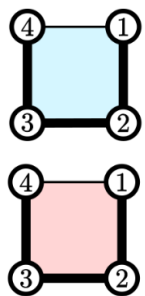
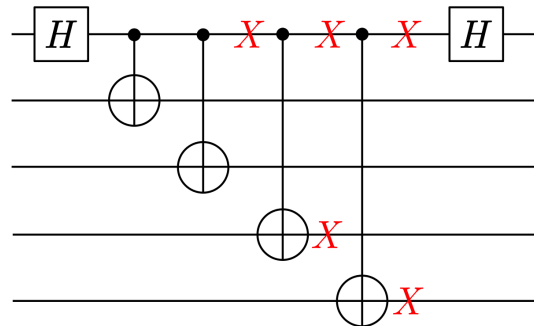
What about error correction?

Maintaining fault-tolerance and code properties imposes additional constraints on circuit design and scheduling.

Example: surface code syndrome measurement

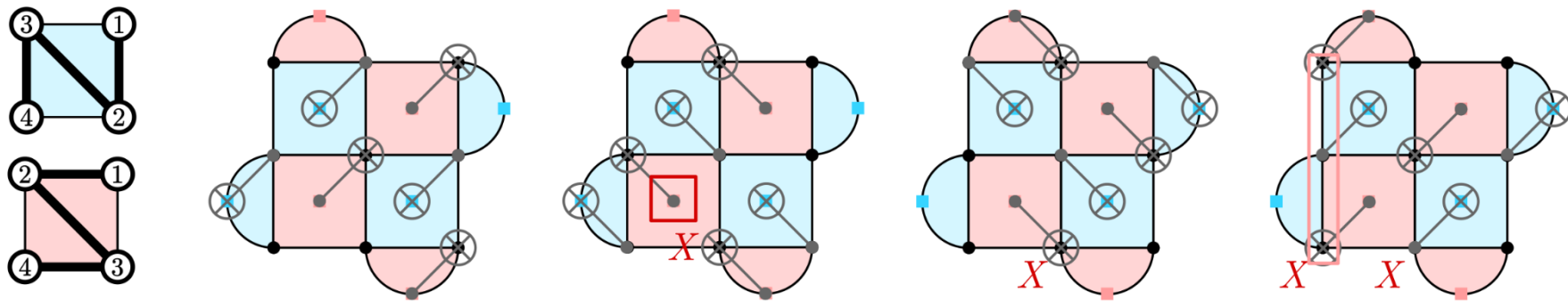


What about error correction?



What about error correction?

This approach works for arbitrary-sized surface codes.



Lots of moving parts

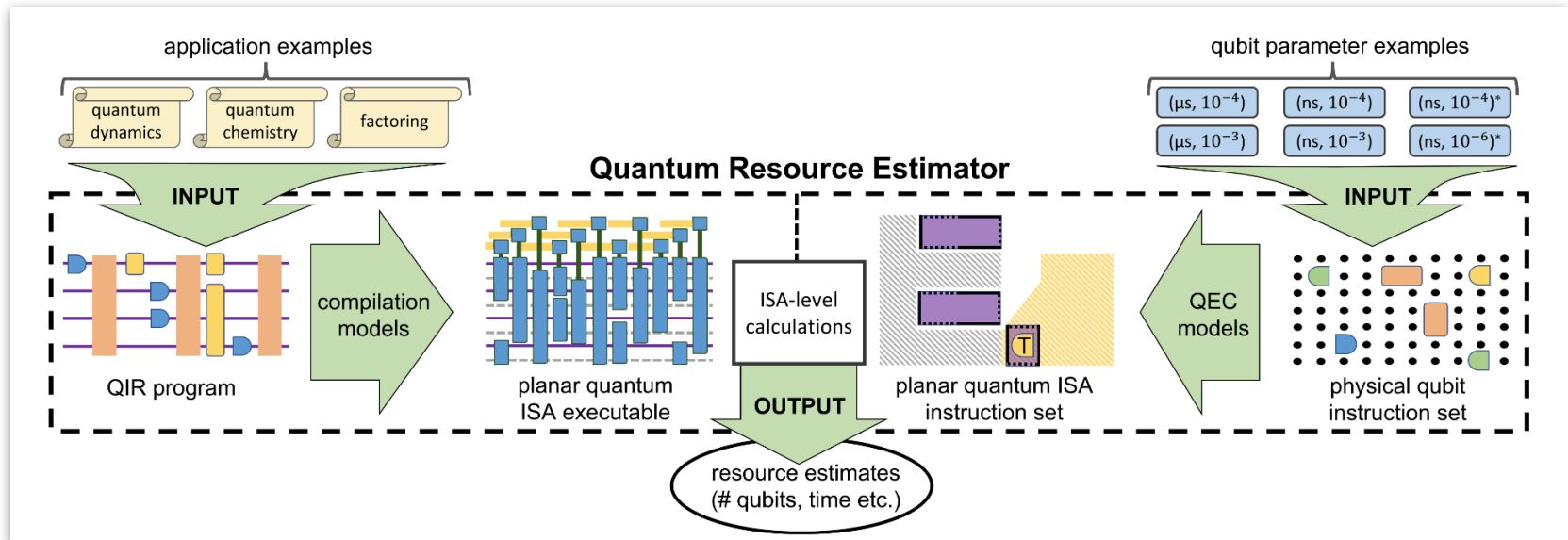
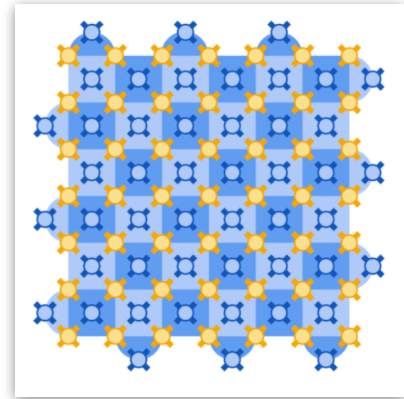


Image (Microsoft Quantum team): M. E. Beverland et al., *Assessing requirements to scale to practical quantum advantage*. arXiv 2211.07629 [quant-ph].

What are we working with?

Resource estimates depend on choice of QECC and hardware features.

- How many qubits?
- What is their coherence time?
- How are they connected?
- How good are 1- and 2-qubit operations?
- How good is readout?
- How long do the operations take?
- What are the main error mechanisms?



Must combine these with algorithmic requirements.

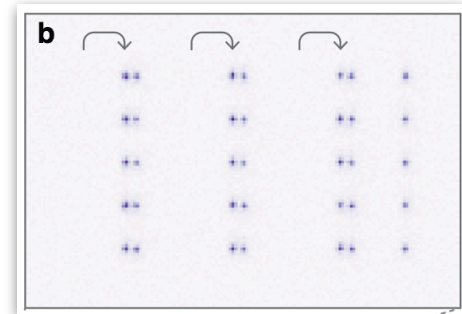
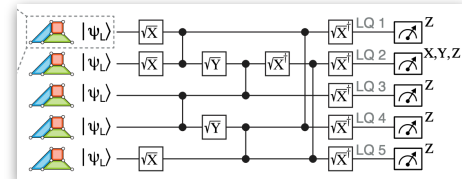
Image: Google Willow processor, <https://research.google/blog/making-quantum-error-correction-work/>

What are we working with?

Make estimates based on recent hardware demos of error correction.

Neutral atoms

- Typical error rates: $\sim 10^{-4}$ 1Q, $\sim 10^{-3}$ 2Q, $\sim 10^{-2}$ readout
- All-to-all connectivity (suitable for qLDPC codes)
- Compilation: shuttling / scheduling
- Relatively slow operation time
 - Coherence times $\sim s$
 - Gate times $\sim \mu s$; spacing required between gates
 - Measurement times $\sim 10s$ of μs
 - Surface code syndrome measurements on order of $\sim ms$



QuEra experiment (^{87}Rb). Magic state distillation, distance 3 colour code. $\sim 99.4\%$ logical fidelity reported.

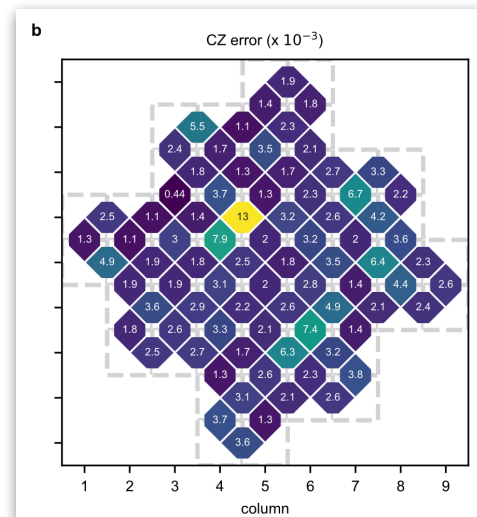
Image source: Sales Rodriguez, P., et al. Experimental demonstration of logical magic state distillation. *Nature* 645, 620–625 (2025).
Metrics: Bluvstein, D., et al. A fault-tolerant neutral-atom architecture for universal quantum computation. *Nature* 649, 39–46 (2026).

What are we working with?

Make estimates based on recent hardware demos of error correction.

Superconducting qubits

- Typical error rates: $\sim 10^{-4}$ 1Q, $\sim 10^{-3}$ 2Q, $\sim 10^{-2}$ readout
- 2D connectivity (suitable for planar codes)
- Compilation: placement, routing (topology-restricted)
- Fast operation time
 - Coherence times ~ 100 s of μ s
 - Gate times ~ 10 s to ~ 100 s of ns
 - Surface code syndrome measurements on order of $\sim \mu$ s



Example CZ error rates from Google's Willow chip. Surface code with $d=7$ achieved 0.143% error per QEC cycle.

Image source: Google Quantum AI and Collaborators. Quantum error correction below the surface code threshold. *Nature* 638, 920-926
Metrics: IBM Quantum Platform;

What are we working with?

Optimistic and pessimistic assumptions (based on surface code)

	Optimistic	Pessimistic
Gate error rate	10^{-4}	10^{-3}
Meas./reset error rate	10^{-3}	10^{-2}
Gate time	100 ns	50 μ s
Measurement time	500 ns	500 μ s

What are we working with?

Context:

- Surface code with lattice surgery
- Magic state distillation with 15-to-1 quantum Reed Muller code
- Circuit-level noise
- *Assume perfect decoder*

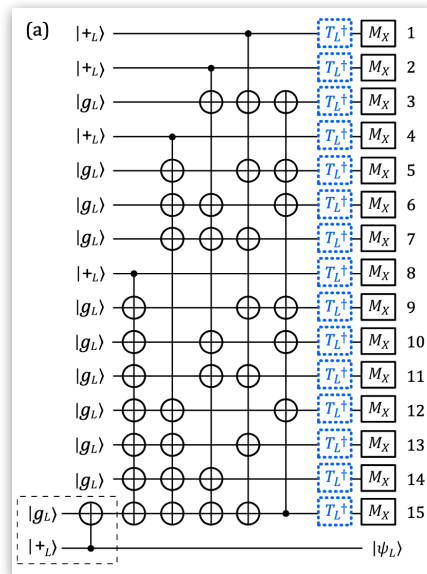
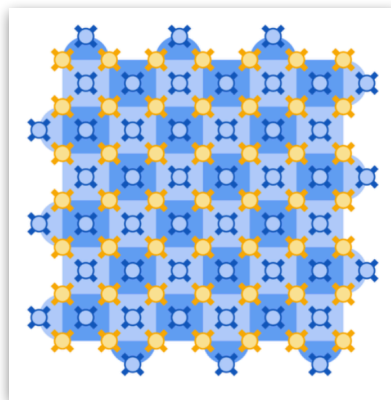


Image: Google Willow processor, <https://research.google/blog/making-quantum-error-correction-work/>

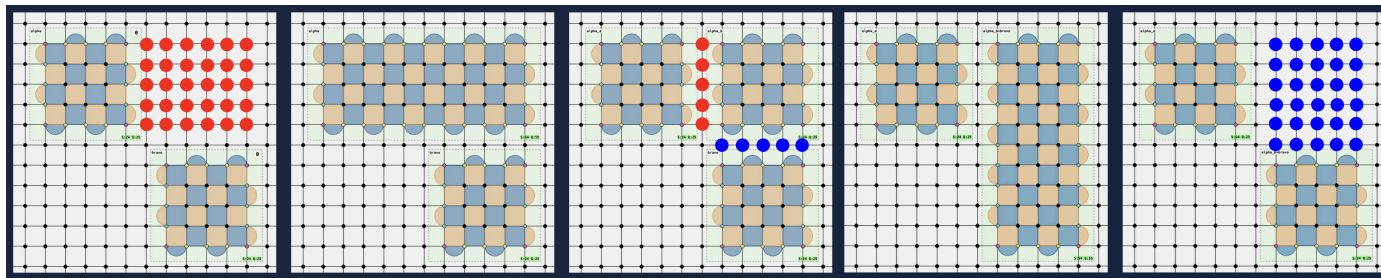
Image: A. G. Fowler, M. Mariantoni, J. M. Martinis, A. N. Cleland. Surface codes: Towards practical large-scale quantum computation. PRA 86, 032324

How good does it need to be?

Lattice surgery instructions for surface code:

- Track logical Paulis through stabilizer evolution
- Operations: Measure X, Measure Z, rotate, grow, shrink, merge, split
- Distance d surface code: d syndrome measurements per cycle

Example: logical CNOT



Images: Entropica Labs, <https://challenge2025.entropicalabs.com/tasks/lattice-surgery-basics-entropica-labs-v250514.pdf>

How good does it need to be?

Consider our Grover implementation

	1 iteration		$\lfloor \pi/4\sqrt{2^N} \rfloor$ iterations	
N	Clifford	T	Clifford	T
4	142	96	426	288
12	654	480	3.27×10^4	2.4×10^4
24	1422	1056	4.57×10^6	3.4×10^6

- Total number of operations: $(4.57 + 3.4) \times 10^6 = 7.97 \times 10^6 \sim 10^7$
- Each Clifford and each T should have error probability less than 10^{-7}

How good does it need to be?

Logical error rate of surface code estimated using:

$$p_L(d) = A \left(\frac{p}{p_{th}} \right)^{\frac{d+1}{2}}$$

Assume $A = 0.03$, $p_{th} = 0.01$ (these come from numerical simulations).

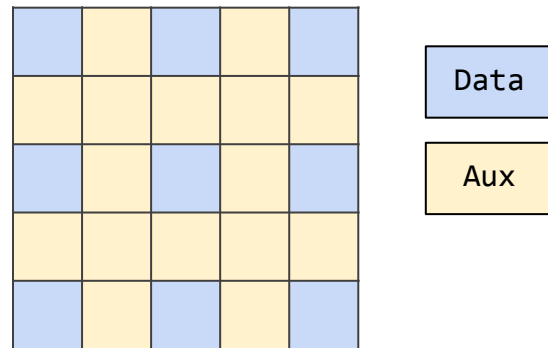
	d	Logical qubits (N)	Physical qubits: $(2d^2 - 1)N$
Optimistic ($p = 10^{-4}$)	5	51	2499
Pessimistic ($p = 10^{-3}$)	11	51	12291

See more detailed analysis: M. E. Beverland et al., *Assessing requirements to scale to practical quantum advantage*. arXiv 2211.07629 [quant-ph].
A. G. Fowler, M. Mariantoni, J. M. Martinis, A. N. Cleland. Surface codes: Towards practical large-scale quantum computation. PRA 86, 032324

How good does it need to be?

We must also assume an architecture with sufficient auxiliary space for CNOTs.

Simplest choice requires 4x overhead.



	d	Logical qubits (N)	Physical qubits: $4(2d^2 - 1)N$
Optimistic ($p = 10^{-4}$)	5	51	9996
Pessimistic ($p = 10^{-3}$)	11	51	49164

See more detailed analysis: M. E. Beverland et al., *Assessing requirements to scale to practical quantum advantage*. arXiv 2211.07629 [quant-ph].
A. G. Fowler, M. Mariantoni, J. M. Martinis, A. N. Cleland. Surface codes: Towards practical large-scale quantum computation. PRA 86, 032324

How long will it take?

	Opt.	Pess.
Gate time (t_g)	100 ns	50 μ s
Meas. time (t_m)	500 ns	500 μ s

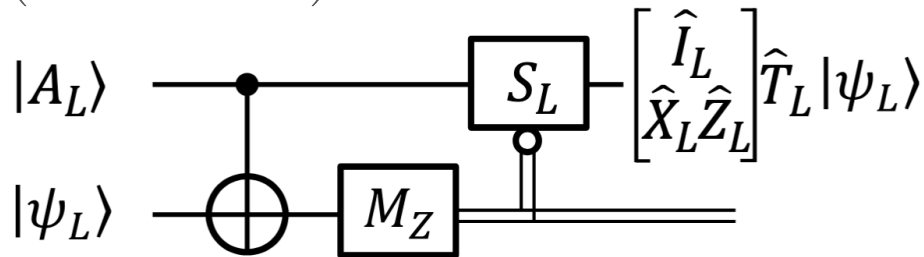
- Assume all Cliffords are CNOT (4.57×10^6)
 - 3 lattice surgery operations (merge/split, which are meas.)
- Cycle time: $t_c = 4 t_g + 2 t_m$
- d measurement cycles per operation
- Total time per operation: $3 d t_c$

	Cycle time (t_c)	d	Wall time
Optimistic	1.4 μ s	5	95.97 s
		11	211.13 s
Pessimistic	1.2 ms	5	22.85 h
		11	50.3 h

How good does it need to be?

Implementing T gates requires **magic states** as resources. Magic states are **injected** and **consumed** to apply logical T .

$$|A_L\rangle = \frac{1}{\sqrt{2}} \left(|0_L\rangle + e^{\frac{i\pi}{4}} |1_L\rangle \right)$$



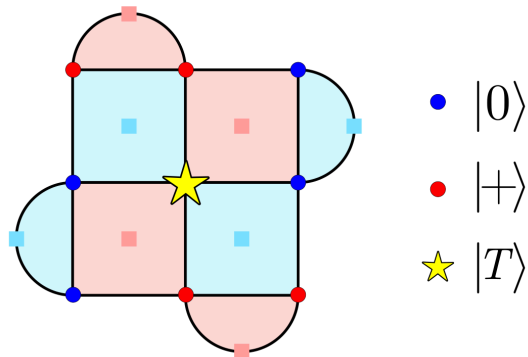
Images: A. G. Fowler, M. Mariantoni, J. M. Martinis, A. N. Cleland. Surface codes: Towards practical large-scale quantum computation. PRA 86, 032324

How good does it need to be?

	Opt.	Pess.
Gate error	10^{-4}	10^{-3}
Reset error	10^{-3}	10^{-2}

Step 1: create a “bad” logical T state

- Initialize qubits
- Two stabilizer measurement cycles
- Post-select on correct outcomes



$$p_L = 9\frac{p_2}{15} + 2\frac{p_1}{3} + p_{reset}$$

	p_L
Opt.	1.1×10^{-3}
Pess.	1.1×10^{-2}

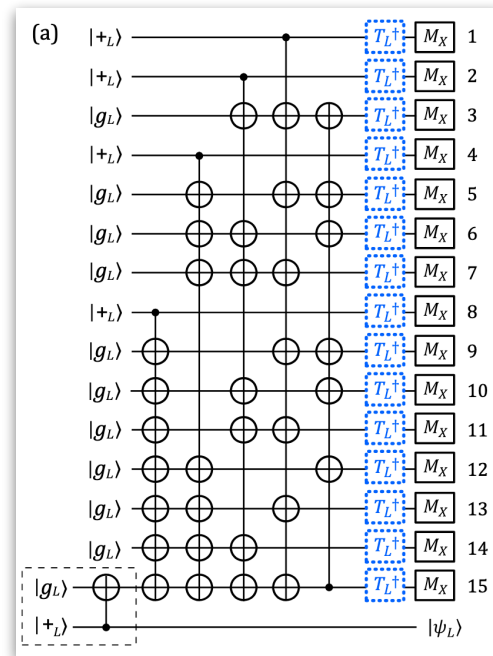
Lingling Lao and Ben Criger. 2022. Magic state injection on the rotated surface code. In Proceedings of the 19th ACM International Conference on Computing Frontiers (CF '22). Association for Computing Machinery, New York, NY, USA, 113-120.

How good does it need to be?

Step 2: use code where T is transversal to **distill** higher-quality states.

Common choice: $[[15, 1, 3]]$ quantum Reed-Muller code.

$$p_o = 35p_i^3$$



Images: A. G. Fowler, M. Mariantoni, J. M. Martinis, A. N. Cleland. Surface codes: Towards practical large-scale quantum computation. PRA 86, 032324

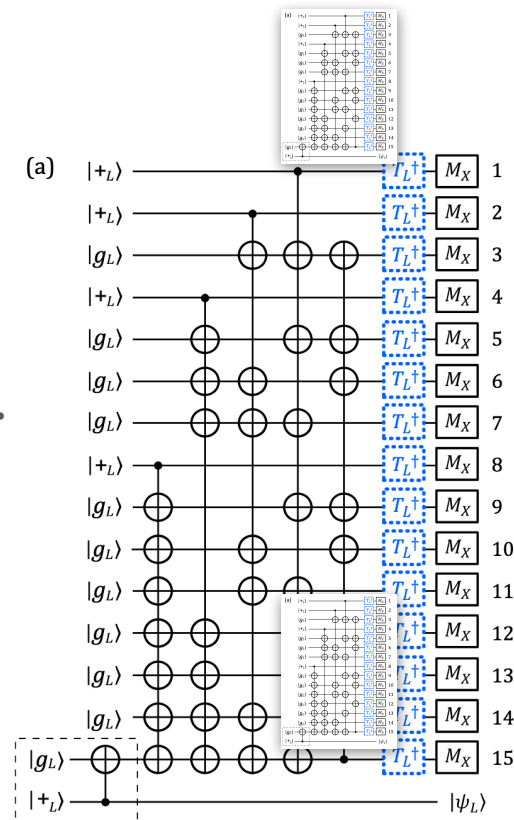
How good does it need to be?

Succeeds with probability $1 - 15p$.

Logical error probability must be $< 10^{-7}$:
may need to concatenate distillation circuits.

$$p_o = 35p_i^3$$

	p_i	p_o	p_o'
Opt.	1.1×10^{-3}	4.66×10^{-8}	
Pess.	1.1×10^{-2}	4.66×10^{-5}	3.54×10^{-12}



See more detailed analysis: M. E. Beverland et al., *Assessing requirements to scale to practical quantum advantage*. arXiv 2211.07629 [quant-ph].
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How good does it need to be?

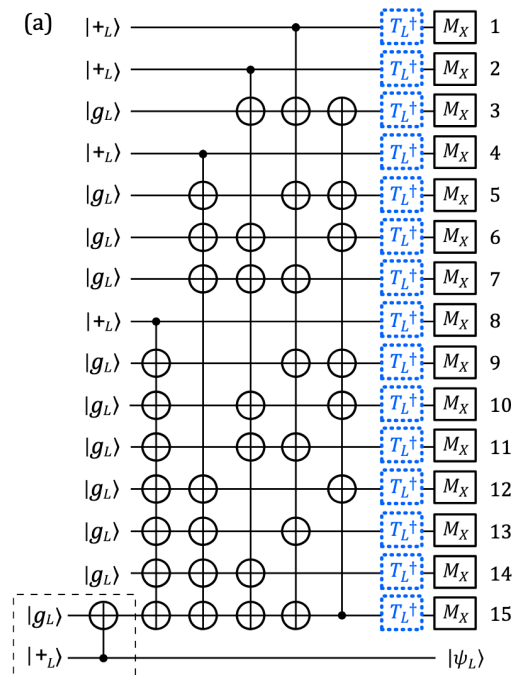
Optimistic: SC with logical error prob p_L and

- 16 logical qubits
- 3 types of Pauli errors
- Est. $4 \times 8 \times 3 \times d$ operations

Must find d s.t.

$$p_L(d) = A \left(\frac{p}{p_{th}} \right)^{\frac{d+1}{2}}, \quad 4608 d p_L(d) < 10^{-7}$$

p_g	d	p_L	$4608 d p_L$
10^{-4}	11	3×10^{-14}	1.52×10^{-9}



See more detailed analysis: M. E. Beverland et al., *Assessing requirements to scale to practical quantum advantage*. arXiv 2211.07629 [quant-ph].
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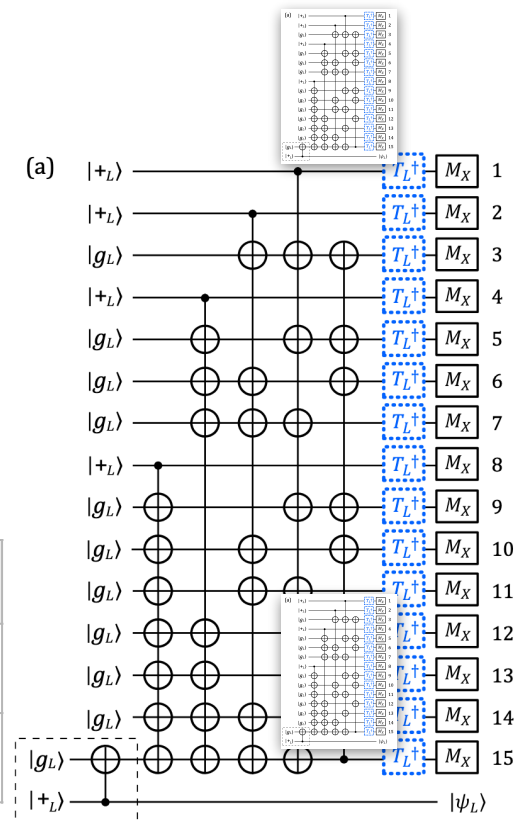
How good does it need to be?

Pessimistic: for round 2, logical error prob p_L and

- 16 logical qubits
- 3 types of Pauli errors
- Est. $4 \times 8 \times 3 \times d$ operations

and for round 1, have 15×16 logical qubits.

	p_g	d	p_L	$4608 \ d \ p_L$	$35(15 * 4608 \ d \ p_L)^3$
R2	10^{-3}	21	3×10^{-13}	2.9×10^{-8}	
R1	10^{-3}	15	3×10^{-10}		1.05×10^{-9}



See more detailed analysis: M. E. Beverland et al., *Assessing requirements to scale to practical quantum advantage*. arXiv 2211.07629 [quant-ph].
 A. G. Fowler, M. Mariantoni, J. M. Martinis, A. N. Cleland. Surface codes: Towards practical large-scale quantum computation. PRA 86, 032324

How many qubits / how long?

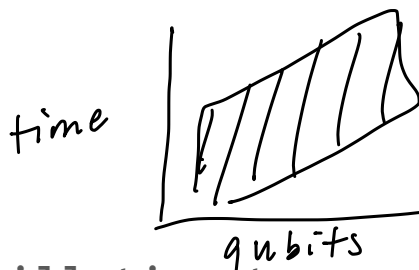
	Opt.	Pess.
Gate time (t_g)	100 ns	50 μ s
Meas. time (t_m)	500 ns	500 μ s

Do a rough estimate:

- 8 layers of CNOTs + 2 measurements per distillation circuit
- Cycle time:
 - $8 \times 3 \times (4 t_g + 2 t_m) \times d$
 - $2 t_m$

	d	Phys. qubits	Time for 1 $ T\rangle$	Total time
Opt.	11	3856	0.37 ms	1257 s
Pess. R1	15	107760	0.43 s	408 h
Pess. R2	21	-	0.61 s	571.2 h

Summary



Incorporate both Cliffords + distillation to compare our best- and worst-case scenarios...

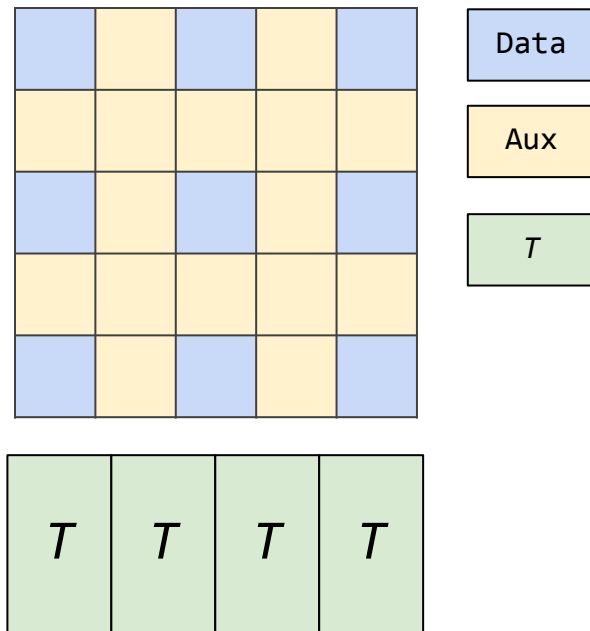
	Parameters	Physical qubits	Wall time (worst case: sequential)
Best	$d = 5$, fast gates 1 round dist. ($d = 11$)	13852	1353 s
Worst	$d = 9$, slow gates 2 rounds dist. ($d = 15, 21$)	156924	1029.5 h (42.9 days)

Limitations

Refine and improve our estimate by...

- Optimizing for T depth
- Parallelizing T state distillation with multiple factories
- Optimizing lattice surgery layout and routing to reduce number of auxiliary regions

Try some of these yourself!

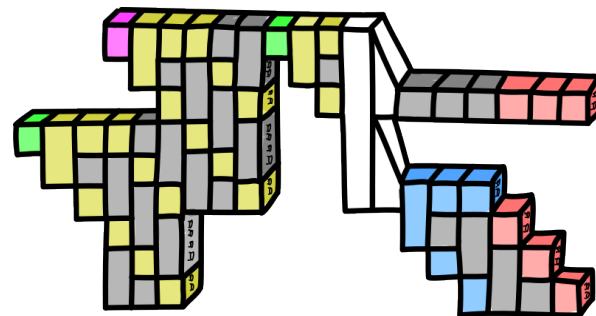
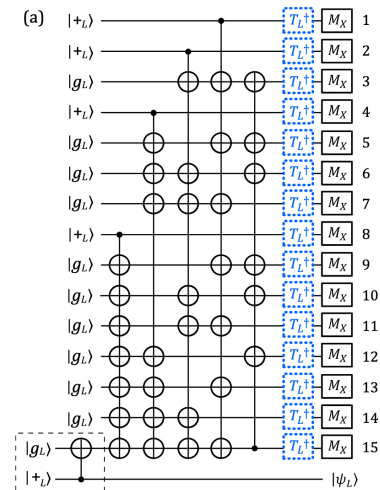
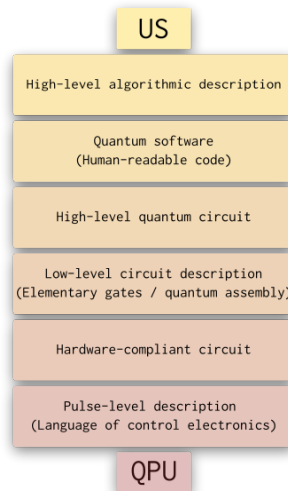


Summary and takeaways

It takes significant effort to determine the **true cost** of running a quantum algorithm.

When performing or assessing a resource estimate, important to carefully consider **assumptions**

- Hardware quality
- Choice of QECC
- Architecture
- Compilation strategies



In case you were curious...

How much does it cost to run a preimage attack on the **SHA2-256** hash function with Grover search on a fault-tolerant quantum computer?

(A) 2128

(B) 2145

(C) 2166

(D) 2127.6



In case you were curious...

Solution “costs” ~275 billion times more than expected from number of Grover queries.

“Cost” =
logical qubits x SC cycles

 [quant-ph](#) > [arXiv:1603.09383](#) Search... Help | Advan

Quantum Physics

[Submitted on 30 Mar 2016 (v1), last revised 30 Nov 2016 (this version, v3)]

Estimating the cost of generic quantum pre-image attacks on SHA-2 and SHA-3

[Matthew Amy](#), [Olivia Di Matteo](#), [Vlad Gheorghiu](#), [Michele Mosca](#), [Alex Parent](#), [John Schanck](#)

		SHA-256	SHA3-256
Grover	T -count	1.27×10^{44}	2.71×10^{44}
	T -depth	3.76×10^{43}	2.31×10^{41}
	Logical qubits	2402	3200
	Surface code distance	43	44
	Physical qubits	1.39×10^7	1.94×10^7
Distilleries	Logical qubits per distillery	3600	3600
	Number of distilleries	1	294
	Surface code distances	{33, 13, 7}	{33, 13, 7}
	Physical qubits	5.54×10^5	1.63×10^8
Total	Logical qubits	$2^{12.6}$	2^{20}
	Surface code cycles	$2^{153.8}$	$2^{146.5}$
	Total cost	$2^{166.4}$	$2^{166.5}$

Table 3. Fault-tolerant resource counts for Grover search of SHA-256 and SHA3-256.

Acknowledgements

- Thanks to the AlgoLab team and school organizers
- Thanks to my PhD student Kilian Poirier for reviewing my slides

Come visit us at UBC!



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